THANOJ BUDDHIMA

Software Engineering Intern
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SUMMARY

Information Technology undergraduate at the University of Moratuwa (CGPA: 3.438/4.00) focused on software engineering, full-stack development, AI, and machine learning. Experienced in developing web applications, backend services, data platforms, and intelligent systems using Java, Python, TypeScript, React, Next.js, NestJS, Spring Boot, and PostgreSQL. Co-Founder and Software Engineer at Gmora, with experience in collaborative product development, technical problem-solving, hackathons, and client-oriented software projects. Seeking an internship to contribute to scalable, secure, and maintainable software solutions.

PROJECTS

Piyoway Tourist Guide (Live) 2026
www.piyoway.com

- Built and deployed a tourism management platform consolidating 4 core services, destination discovery, itinerary planning, reviews, and listings, for 4 user roles, supported by administrative dashboards, cloud-based media management, and a Progressive Web App (PWA) architecture.
- **Stack:** Next.js, TypeScript, NestJS, PostgreSQL, Prisma, Tailwind CSS, Docker, Nginx, Cloudinary

SuwaPath – AI Healthcare Navigation Platform 2026
github.com/Pasidu-Mihiranga/SuwaPath

- Engineered a healthcare platform comprising a collection of specialized ML and vision models, orchestrated by a bounded LangGraph agentic loop and a 23-rule deterministic triage engine to achieve 100% emergency detection (8/8 cases).
- Developed a multilingual NLP pipeline mapping English, Sinhala, and Tamil symptom inputs to ∼90 clinical concepts, enabling capability-aware matching to route patients to appropriate hospital specialties and facilities.
- Integrated predictive models for 7-day specialty demand forecasting and automated medical report OCR, benchmarking the multi-model ecosystem against a 114-vignette gold dataset and passing 150+ automated agentic scenario checks.
- **Stack:** Python, FastAPI, React, TypeScript, LangGraph, PostgreSQL, Qdrant, Tailwind

PlanGmora – Agile Project Management Platform 2026
github.com/ThanojBuddhima/ProjectManagementTool

- Engineered a Jira-inspired full-stack project management platform with 4 project views (Summary, List, Kanban Board, and Timeline) to support project planning, progress tracking, and Agile workflow management.
- Implemented issue tracking across 4 issue types, 5 priority levels, and 4 workflow statuses, integrating drag-and-drop Kanban boards, subtasks, team assignments, and Gantt chart-based scheduling.
- Developed JWT authentication, 4-tier role-based access control (RBAC), email verification, and 2 notification channels (in-app and email) to support secure collaboration, activity tracking, and project-level access management.
- **Stack:** React, TypeScript, NestJS, PostgreSQL, TypeORM, JWT, Vite

Garbo – Smart Waste Management Platform 2026
github.com/CodeMIndsUoM/Garbo_backend | github.com/CodeMIndsUoM/Garbo-flutter
github.com/CodeMIndsUoM/Garbo_web_dashboard

- Developed a smart municipal waste management platform supporting 6 user roles, including citizens, administrators, field mentors, bin collectors, and third-party collectors, with multi-tenant access control and council-level data isolation.
- Implemented automated waste collection route optimization using Google OR-Tools (CVRP/VRP) and OSRM distance matrices, considering 3 routing factors: vehicle capacity, bin priority, and route distance for collection assignment and progress tracking.
- Integrated 3 operational capabilities, real-time updates through WebSockets (STOMP), Firebase Cloud Messaging notifications, and GPS-enabled complaint reporting, alongside citizen gamification, waste collection requests, and third-party recycler bidding workflows.
- **Stack:** Java, Spring Boot, Spring Security, JWT, PostgreSQL, Spring Data JPA, Hibernate, Flyway, Google OR-Tools, OSRM, WebSockets, Firebase Cloud Messaging, Docker

AppLock Pro – AI-Powered macOS Application Locker 2026
github.com/ThanojBuddhima/applockPro

- Engineered a native macOS application security tool using on-device facial recognition to restrict access to selected applications, combining 2 authentication factors/methods: biometric verification and password-based fallback.
- Integrated Apple's Vision framework and Core ML with ArcFace face embeddings to implement facial feature extraction and identity verification, enabling privacy-focused authentication without cloud-based image processing.
- Implemented 3 core access-control features: facial recognition, password fallback verification, and configurable session timeouts, strengthening local application security and user privacy.
- **Stack:** Swift, SwiftUI, Core ML, Apple Vision, ArcFace, macOS

EXPERIENCE

Co-Founder / Software Engineer – Gmora Feb 2026 – Present

- Collaborated with a 9-member team to develop and deliver 3 software projects, contributing to full-stack implementation and digital product development.
- Contributed to software architecture, feature development, and technical decisions for client-focused solutions, using Git-based version control and iterative workflows.
- Worked with teammates on hackathons and software delivery, applying cross-functional problem-solving and continuous improvement.

STEM Intern – Gavesha Labs 2025 – 2026

- Created STEM learning resources and technical content, applying research, communication, and teamwork skills in educational initiatives.

TECHNICAL SKILLS

Programming: Python, Java, C, C++, JavaScript, TypeScript, PHP, SQL, Swift
Web and Backend: React.js, Next.js, Node.js, Express.js, NestJS, Spring Boot, FastAPI, REST APIs, HTML, CSS, Tailwind CSS
Databases: PostgreSQL, MySQL, SQLite, MongoDB, Redis, Firebase, Supabase
AI/ML and Data: PyTorch, Scikit-learn, OpenCV, MediaPipe, LangGraph, Qdrant, Pandas, NumPy, CUDA, Core ML, Apple Vision
Tools and Platforms: Git, GitHub, GitHub Actions, Docker, Nginx, Conda, Selenium, Linux (Ubuntu), Windows
Software Engineering: OOP, Data Structures and Algorithms, Agile Development, Kanban, JWT, RBAC, debugging, software testing
Design Tools: Figma, Canva
Soft Skills: Project Management, Public Speaking, Team Collaboration

EDUCATION

University of Moratuwa 2023 – Present
B.Sc. (Hons) in Information Technology | CGPA: 3.438/4.00 Expected graduation: 2027

Richmond College, Galle 2022
G.C.E. Advanced Level – Physical Science: Combined Mathematics (A), Physics (A), Chemistry (B)

AWARDS AND ACHIEVEMENTS

- **Champion – AI Challenge Sri Lanka** (2026), Team Gmora; multi-agent AI system, organized by IEEE Young Professionals Sri Lanka.
- **Champion – BITCode v6.0** (2026), Team Gmora; coding competition hosted by Rajarata University.
- **Champion – InnovateX 2025**, Team Gmora; enterprise software solution, organized by Zebra Technologies.
- **Champion – OctWave 2.0 (ML Track)** (2025); predictive machine learning models, IEEE Student Branch, University of Moratuwa.
- **2nd Runner-Up – IgniteX by Hutch** (2026), Team Gmora; 3rd place in the competition organized by Hutch Sri Lanka.
- **1st Runner-Up – Cre8x 3.0** (2026), Team Gmora; National Level Inter-University UI/UX Designathon organized by the BCS Student Chapter, KDU; 8-hour design competition.

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification – DeepLearning.AI (2026)
  coursera.org/share/3992ebf0e0ca2561a53484e38cb617bd
- Intro to Deep Learning – Kaggle
  kaggle.com/learn/certification/thanojbuddhima/intro-to-deep-learning
- Pandas – Kaggle
  kaggle.com/learn/certification/thanojbuddhima/pandas
- AI Fluency for Educators – Anthropic
  verify.skilljar.com/c/cjzmcggamfh3

VOLUNTEERING

Mathematics Olympiad Training – Southern Province 2024, 2025
Tutored and mentored students in mathematics Olympiad training.

Sasnaka Sansada 2023 – 2024
Contributed to the Ganitha Saviya initiative and mathematics seminars reaching students across 40+ schools in Sri Lanka.

REFERENCES

K. A. Dilini T. Kulawansa
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